

# Francesco Mazzei

*Curriculum Vitae*


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## ABOUT ME

Ph.D. candidate conducting research at the EP-DT-EO section at CERN, specializing in robotics and automation systems for harsh indoor environments. My doctoral research within the EP-R&D program, applied inside the ALICE experimental cavern, has convinced me that robotics and autonomous systems will play an increasingly central role in supporting large-scale scientific infrastructure, from simple assembly and disassembly tasks to operations carried out in close collaboration with human personnel. I am now seeking to continue this path through a postdoctoral position at CERN, contributing to the development of autonomous inspection and monitoring systems for the experimental caverns of future collider programs, including the Future Circular Collider (FCC).

## WORK EXPERIENCE

### *Spacecraft Payload System Engineer*

National Institute for Astrophysics (INAF)
MAR 2023 – OCT 2023 (ROME, ITALY)

- Managed and maintained the MBSE model for the ESA PLATO mission cameras within the Camera System Engineering team, ensuring full requirements traceability.
- Ensured compliance with system-level requirements throughout camera integration and calibration campaigns, identifying, tracking, and resolving technical deviations to support key project milestones.
- Authored and revised technical documentation, including requirement specifications and interface control documents, supporting formal review processes with ESA.

### *Graduate Robotics Engineer*

European Organization for Nuclear Research (CERN)
JUN 2022 – NOV 2022 (GENEVA, SWITZERLAND)

- Conducted research within the EP-DT-EO section under the EP-R&D program, in collaboration with the School of Aerospace Engineering in Rome, Italy, developing robust control systems for aerial robots intended for deployment in experimental caverns at CERN.
- Validated the actuation system within the EP-DT-DI section, characterizing the impact of high magnetic fields on DC motor performance through a dedicated test campaign.
- Supervised a Bachelor student on a project investigating the optimal placement of cameras within the motion capture system used for robot localization.
- Contributed to a human motion capture study within the EP-DT-SD section, in collaboration with the University of Sheffield, supporting gesture recognition and radiation dose estimation for personnel exposure in high-radiation areas of the experimental cavern.

### *MSc. Thesis Internship in Robotics*

European Organization for Nuclear Research (CERN)
NOV 2021 – DEC 2021 (GENEVA, SWITZERLAND)

- Carried out my Master thesis research within the EP-DT-EO section under the EP-R&D program, conducting a feasibility study on deploying a lightweight blimp robot inside experimental caverns for the CERN FCC.

## COMPUTER SKILLS

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|---|---|
| PROGRAMMING | Matlab & Simulink, Python, C++ |
| HARDWARE | Arduino, Raspberry Pi, ESP32 |
| SOFTWARE | KiCad, CATIA, Unreal Engine |
| WRITING | LaTeX, MS Office |

## COMMUNICATION SKILLS

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|---|---|
| CONFERENCES | Oral presentations at international conferences, workshop and seminars (ERF 2022, IAC 2023, AAIC 2024) |
| POSTERS | Poster presentations, EP-R&D Day, 2024–2026, available at CERN CDS |

## MANAGERIAL SKILLS

- During my PhD at CERN I supervised and mentored 5+ Bachelor's and Master's students , guiding their thesis projects from goal definition to final write-up.
- During my PhD at CERN I was responsible for major technical decisions, from system design to materials procurement, overseeing budget allocation for the project.

## AWARDS

2023 **"Antonio Ventura" Scholarship**
Awarded with a scholarship of €2000 for the best Master's thesis in Space and Astronautical Engineering at the University of Rome "La Sapienza" by the "Noi Sapienza" Foundation.

2023 **"Excellent Graduate"**
Awarded as an "Excellent Graduate" of the year 2020/2021 in Space and Astronautical Engineering during the 10th Graduate Day by the "Noi Sapienza" Foundation of University of Rome "La Sapienza".

## CERTIFICATIONS (ECTS/CPE)

- **SpaceRaise 2026 – Robotics for Space**
  Gran Sasso Science Institute (GSSI), May 2026
- **Robotic Vision Summer School 2025**
  The Australian National University, Feb 2025

- Developed the nonlinear dynamic model of an indoor airship and characterized through simulations the magnetic disturbances on its electric motors from the magnetic field generated by a future FCC particle detector.
- Designed nonlinear controllers for trajectory tracking and validated them experimentally using a blimp platform equipped with a motion-capture system, benchmarking flight test results against simulation predictions.

## EDUCATION

### PhD student in Space Science and Techniques

*University of Rome Tor Vergata*
*European Organization for Nuclear Research (CERN)*
2023 – 2026 (GENEVA, SWITZERLAND)

- My doctoral research, funded by CERN within the EP-R&D program, focused on the design, development, and experimental validation of an autonomous aerial robotic platform for 3D mapping and inspection inside the ALICE experimantal cavern.
- Technical contributions spanned embedded real-time control systems, computer-vision localization, sensor data acquisition, data analysis, CAD-based mechanical design, 3D printing, and custom PCB design.
- Conducted experimental test campaigns inside the LHC cavern, with a focus on the ALICE experiment, during official technical stops and magnet ramp-up periods, in collaboration with the ALICE Run Coordination.
- Authored peer-reviewed journal articles and conference papers, presented research findings at international conferences and workshops, and supervised Bachelor's and Master's students as thesis advisor.

### MSc. in Space and Astronautical Engineering

*University of Rome La Sapienza*
110/110 Cum Laude
2020 – 2021 (ROME, ITALY)

- Built a solid foundation in flight dynamics, control theory, spacecraft systems and mission design, propulsion, and space robotics, culminating in an experimental thesis project conducted in collaboration with CERN.
- Title of thesis: "Dynamic interaction and control of an indoor blimp inside the CERN FCC-hh magnetic environment"

### BSc. in Aerospace Engineering

*University of Rome La Sapienza*
110/110 Cum Laude
2016 – 2019 (ROME, ITALY)

- Built a solid foundation in mathematics, physics, and engineering mechanics, followed by aerospace-specific coursework in aerodynamics, flight mechanics, propulsion, and space systems, culminating in a thesis on space environment.
- Title of thesis: "Doses of radiation from solar protons absorbed for a trip to Mars"

- **IEEE Multi-Robot Systems Summer School**
  Czech Technical University in Prague, Jul 2024
- **Clean Space Training Course 2024**
  European Space Agency (ESA), Mar 2024
- **Space Studies Program (SH-SSP)**
  International Space University (ISU)
  University of South Australia (UniSA), 2023
- **Astrobiology: Exploring Other Worlds**
  University of Arizona, Mar 2023
- **5+ MATLAB/Simulink Onramp Courses**
  MathWorks, 2022
- **10+ Python & Data Science Certifications**
  Madecraft, Sololearn, 2022

## PUBLICATIONS

F. Mazzei et al., "Experimental Characterization of Miniature DC Motors for Robotics in High Magnetic Field Environments," *Robotics*, vol. 14, no. 12, p. 172, 2025.

F. Mazzei et al., "Development of Small Airships for Indoor Exploration and Inspection of the CERN Underground Facilities," in *Proceedings of the Airship Association International Conference (AAIC)*, 2024.

F. Mazzei et al., "Mission analysis, dynamics and robust control of an indoor blimp in a CERN detector magnetic environment," *Robotics and AI*, vol. 10, p. 1 238 081, 2023, ISSN: 2296-9144.

F. Mazzei et al., "Technological and Legal Perspectives for Sustainable Human Presence on the Moon," in *Proceedings of the International Astronautical Congress (IAC)*, 2023.

For a complete and updated list of publications, including co-authored works, please refer to my ORCID profile: `https://orcid.org/0000-0003-4888-1428`

## LANGUAGE

**Italian** ●●●●●
**English** ●●●●●
**French** ●●●○○
**Spanish** ●○○○○

## MY INTERESTS

- Musician (drummer & bassist)
- Audio and video editing for music production using DAWs, NLEs
- Sports (running, swimming and football)
- Traveling and outdoor activities